
LOCALLY DECODABLE CODES

SAYAN KAR

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Abstract

Locally Decodable Codes as discussed in [[Yek12](#)].

NOTATIONS

$\mathbb{F}_q$	Finite field of size q
$\mathbb{F}_q^*$	Multiplicative group of $\mathbb{F}_q$
$\mathbb{Z}_m$	residue class of m
$\mathbb{N}$	Set of natural numbers
$[n]$	The set $\{1, 2, \dots, n\}$
$\emptyset$	Empty set
$X \sim \mathcal{D}$	Random variable X having distribution $\mathcal{D}$
$\Delta(\cdot, \cdot)$	Relative Hamming distance
$\mathbb{E}[X]$	Expectation of random variable X
$\mathbb{D}[X]$	Variance of random variable X
$\mathbf{w}(l)$	l -th coordinate of $\mathbf{w} \in \mathbb{F}_q^n$
D -evaluation of h over D	Vector of values of h at all points of D
$\text{Unif}(S)$	Uniform distribution over the set S
q	A prime power, unless stated otherwise
$g^{\mathbf{w}}$	$(g^{\mathbf{w}(1)}, \dots, g^{\mathbf{w}(n)})$
$M_{\mathbf{w}, \mathbf{v}}$	$\{g^{\mathbf{w} + \lambda \mathbf{v}} \mid \lambda \in \mathbb{Z}_m\}$
$\text{mon}_{\mathbf{u}}(z_1, \dots, z_n)$	$\prod_{\ell} z_{\ell}^{\mathbf{u}(\ell)}$ in $\mathbb{F}_q[z_1, \dots, z_n] / (z_1^m - 1, \dots, z_n^m - 1)$

$\langle \mathbf{u}, \mathbf{v} \rangle$	Inner product of vectors $\mathbf{u}$ and $\mathbf{v}$
$\text{supp}(h(y))$	$\{e \in \mathbb{N}_{\geq 0} : y^e \text{ has positive coefficient in } h\}$
b -bounded set	Every element of the set ($\subseteq \mathbb{R}$) is less than b
$\text{agr}(e, M)$	Number of agreements of M -evaluation of F and C_m -evaluation of h
$\text{weight}(e, M)$	$\max\{\text{agr}(e, M) : h \in \mathbb{F}[y], h(0) = e, \text{supp}(h) \subseteq S \cup \{0\}\}$
$A \otimes B$	Kronecker product of A and B
$A \odot B$	Hadamard Product of A and B
$\ \mathbf{a}\ $	Hamming weight of $\mathbf{a}$

CONTENTS

Notations	ii
1 Introduction	1
1.1 Definition of Local Decoding and Correction	1
1.2 Reed Muller locally decodable codes	2
1.2.1 Basic Linear Local Correction	2
1.2.2 Improvement on line local decoding	3
1.2.3 Decoding on Curves	4
1.3 Binary Code Correction	5
2 Matching Vector Codes	8
2.1 Introduction	8
2.2 Some Decoding Procedures	10
2.2.1 Basic Decoding on Lines	10
2.2.2 Improvement on Line Decoding	11
2.2.3 Decoding on Multiple Lines	12
3 Matching Vectors	14
3.1 Degree Analysis of OR_n Function	14
3.1.1 Polynomial Representation of OR-function in mod 6	15
3.1.2 Barrington-Beigel-Rudich Theorem	16
3.2 Construction of MVF	18

4	Private Information Retrieval	20
4.1	Introduction	20
4.2	A Basic PIR Scheme	21
4.3	PIR Scheme of [GKS25]	22
4.3.1	Preliminaries	23
4.3.2	0-interpolation Property	25
4.3.3	The PIR Scheme	25
A	Appendix	29

CHAPTER 1

INTRODUCTION

1.1 Definition of Local Decoding and Correction

This chapter mostly consists of *local correction* of some linear codes *along some curves*, mainly linear and quadratic. Let us first define what is local correction and decoding formally.

Definition 1.1 (Locally Decodable Code). A q -ary code $C : \mathbb{F}_q^k \rightarrow \mathbb{F}_q^N$ is said to be (r, δ, ε) -*locally decodable* if there exists a randomized decoding algorithm $\mathcal{A}$ such that

1. For all $\mathbf{x} \in \mathbb{F}_q^k$, $i \in [k]$, and all vectors $\mathbf{y} \in \mathbb{F}_q^N$ satisfying $\Delta(C(\mathbf{x}), \mathbf{y}) \leq \delta$, we have:

$$\Pr[\mathcal{A}^{\mathbf{y}}(i) = \mathbf{x}(i)] \geq 1 - \varepsilon,$$

where the probability is taken over the random coin tosses of the algorithm $\mathcal{A}$.

2. $\mathcal{A}$ makes at most r queries to $\mathbf{y}$.

Definition 1.2 (Locally Correctable Code). A code $C \subseteq \mathbb{F}_q^N$ is said to be (r, δ, ε) -*locally correctable* if there exists a randomized correcting algorithm $\mathcal{A}$ such that

1. For all $\mathbf{c} \in C$, $i \in [N]$, and all vectors $\mathbf{y} \in \mathbb{F}_q^N$ satisfying $\Delta(\mathbf{c}, \mathbf{y}) \leq \delta$, we have

$$\Pr[\mathcal{A}^{\mathbf{y}}(i) = \mathbf{c}(i)] \geq 1 - \varepsilon,$$

where the probability is taken over the random coin tosses of the algorithm $\mathcal{A}$.

2. $\mathcal{A}$ makes at most r queries to $\mathbf{y}$.

The main difference between them is that in locally decodable code, we want to retrieve coordinates of the sent message from recieved word and for locally correctable codes, we want to retrieve the coordinates of codewords. Lemma 2.3 in [Yek12] shows that a linear locally correctable code implies existence of a locally decodable code with the same parameter as of the first code.

1.2 Reed Muller locally decodable codes

Before going into the correction of Reed Muller, recall the definition of Reed Muller code.

Definition 1.3 (Reed Muller Codes). For a finite field $\mathbb{F}_q$, define evaluation of $f \in \mathbb{F}[x_1, \dots, x_n]$ at $\mathbf{z} = (z_1, \dots, z_n) \in \mathbb{F}_q^n$ to be

$$\text{Eval}_{\mathbf{z}}(f) = f(z_1, \dots, z_n).$$

Also define $\text{Eval}(f) := (\text{Eval}_{\mathbf{z}_1}(f), \dots, \text{Eval}_{\mathbf{z}_{q^n}}(f)) \in \mathbb{F}_{q^n}$ for some fixed ordering in $\mathbb{F}_{q^n}$. Now define *Reed-Muller Code* of order r as

$$\text{RM}(n, r) := \{\text{Eval}(f) : f \in \mathbb{F}[x_1, \dots, x_n], \deg f \leq r\}.$$

Our main goal of this section is to recover the evaluation of $f \in \mathbb{F}_q[z_1, \dots, z_n]$ at $\mathbf{w} \in \mathbb{F}_q^n$ from evaluations, possibly corrupted, of $f \in \mathbb{F}_q[z_1, \dots, z_n]$ at some points on a curve passing through $\mathbf{w}$. So for the most basic local correction, we take a line passing through $\mathbf{w}$ first, then move into degree 2 curve.

1.2.1 Basic Linear Local Correction

The main proposition governing this section is the following.

Proposition 1.4. *Let n and d be positive integers. Let q be a prime power, $d < q - 1$; then there exists a linear code of dimension $k = \binom{n+d}{d}$ in $\mathbb{F}_q^N$, $N = q^n$, that is $(d+1, \delta, (d+1)\delta)$ -locally correctable for all δ .*

We shall find the failure probability, i.e., the ε among the three parameters. The corrector procedure is given fully in detail in Proposition 2.4 in [Yek12]. The main idea of the proof is

to look at all $\mathbb{F}_q^n$ evaluations of a total degree d multivariate polynomial $f \in \mathbb{F}_q[z_1, \dots, z_n]$. The goal of the linear corrector would be to retrieve $f(\mathbf{w})$ for a given $\mathbf{w} \in \mathbb{F}_q^n$, with only the recieved word $\tilde{\mathbf{c}}$ with high probability. Now what the corrector does is, it takes $d + 1$ random points(uniform) on a random line passing through $\mathbf{w}$ and queries $\tilde{\mathbf{c}}$ at those evaluation points and find a univariate polynomial h of degree atmost d that is the restriction of f to that line. We shall show that the failure probability of the corrector is atmost $(d + 1)\delta$.

Proof of failure probability. Consider $\mathbf{v} \sim \text{Unif}(\mathbb{F}_q^n)$ and the line $L = \{\mathbf{w} + \lambda \mathbf{v} : \lambda \in \mathbb{F}_q\}$. Also consider a uniformly sampled subset S of $\mathbb{F}_q^*$ of size $d + 1$. Now as the corrector queries the points, we observe that for all $\lambda \in S$,

$$\Pr[f(\mathbf{w} + \lambda \mathbf{v}) \neq c_{\mathbf{w} + \lambda \mathbf{v}}] \leq \delta.$$

Now the success probability is

$$\begin{aligned} \Pr[f(\mathbf{w} + \lambda \mathbf{v}) = c_{\mathbf{w} + \lambda \mathbf{v}} \text{ for all } \lambda \in S] &= 1 - \Pr[f(\mathbf{w} + \lambda \mathbf{v}) \neq c_{\mathbf{w} + \lambda \mathbf{v}} \text{ for some } \lambda \in S] \\ &\geq 1 - \sum_{\lambda \in S} \Pr[f(\mathbf{w} + \lambda \mathbf{v}) \neq c_{\mathbf{w} + \lambda \mathbf{v}}] \\ &\geq 1 - (d + 1)\delta. \end{aligned}$$

□

1.2.2 Improvement on line local decoding

Now naturally we would want to decrease the failure probability, thus we would want to make the above decoding better. Here, keep the code same and try to interpolate the univariate restriction polynomial from fewer query. For this, we would need d to be very smaller than q . The main proposition for this section is the following:

Proposition 1.5. *Let $0 < \sigma < 1$, n and d be positive integers. Let q be a prime power such that $d \leq \sigma(q - 1) - 1$; then there exists a linear code of dimension $k = \binom{n+d}{d}$ in $\mathbb{F}_q^N$, $N = q^n$, that is $(q - 1, \delta, 2\delta/(1 - \sigma))$ -locally correctable for all δ .*

We use basically the same setup as before but change the interpolation part, here we query on atmost $(1 - \sigma)(q - 1)/2$ points and interpolate the polynomial via *Berlekamp-Welch* as given in [GRS25]. The reason for such a weird query number is because beyond this, Berlekamp-Welch can not uniquely interpolate the polynomial. As $d < \sigma(q - 1)$, we have $(1 - \sigma)(q - 1)/2 > (q - 1 - d)/2$, this surpasses the RS bound for unique decoding.

Proof of failure probability. We shall calculate the failure a bit differently here, we shall do it via the probability of the random variable X denoting the number of corrupted query points is very big, i.e., $\Pr[X \geq (1 - \sigma)(q - 1)/2]$. This we can get an upper bound by *Markov Inequality* as given in [AS00] if we can find $\mathbb{E}[X]$. If we can find the expectation, the rest is just calculations that is done in the proof of Proposition 2.5 in [Yek12].

Notice that

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E} \left[\sum_{\lambda \in \mathbb{F}_q^*} \mathbb{1}_{\{f(\mathbf{w} + \lambda \mathbf{v}) \text{ is corrupted}\}} \right] \\ &= \sum_{\lambda \in \mathbb{F}_q^*} \Pr[f(\mathbf{w} + \lambda \mathbf{v}) \text{ is corrupted}] \\ &\leq \sum_{\lambda \in \mathbb{F}_q^*} \delta = (q - 1)\delta \end{aligned} \tag{1.1}$$

This now gives the failure probability for the worst possible case:

$$\begin{aligned} \Pr[X \geq (1 - \sigma)(q - 1)/2] &\leq \frac{\mathbb{E}[X]}{(1 - \sigma)(q - 1)/2} \\ &\leq \frac{2\delta}{1 - \sigma}. \end{aligned}$$

□

1.2.3 Decoding on Curves

Now we go from the line to a quadratic parametrized curve, i.e., we consider the curve $\chi = \{\mathbf{w} + \lambda \mathbf{v}_1 + \lambda^2 \mathbf{v}_2 : \lambda \in \mathbb{F}_q^*\}$, where $\mathbf{v}_1, \mathbf{v}_2 \sim \text{Unif}(\mathbb{F}_q^n)$.

Proposition 1.6. *Let $0 < \sigma < 1$, n and d be positive integers. Let q be a prime power such that $d \leq \sigma(q-1) - 1$; then there exists a linear code of dimension $k = \binom{n+d}{d}$ in $\mathbb{F}_q^N$, $N = q^n$, that for all positive $\delta < 1/2 - \sigma$ is $(q-1, \delta, O_{\sigma, \delta}(1/q))$ -locally correctable.*

Here we shall take the same code and everything would be the same as Proposition 1.5 but the only difference would be that the interpolating polynomial is having degree atmost $2d$ and number of query points is atmost $(1 - 2\sigma)(q-1)/2$. We shall calculate the failure probability by the same random variable here also but in a different way, and instead of using the Markov inequality, we shall be using Chebyshev Inequality, given in [AS00].

Proof of failure probability. We want to find the failure probability, that is

$$\Pr \left[X \geq \frac{1}{2}(1 - 2\sigma)(q-1) \right].$$

Notice that the indicator random variables in 1.1 are independent, thus the variance of X becomes $\mathbb{D}[X] \leq (\delta - \delta^2)(q-1)$. So by applying Chebyshev, we get failure probability in worst case scenario.

$$\begin{aligned} \Pr \left[X \geq \frac{1}{2}(1 - 2\sigma)(q-1) \right] &\leq \frac{\mathbb{D}[X]}{((1 - 2\sigma)(q-1)/2 - (q-1)\delta)^2} \\ &= O_{\delta, \sigma}(1/q). \end{aligned}$$

□

1.3 Binary Code Correction

In this section, we shall use whatever idea we have developed so far and use Code Concatenation, given in [GRS25], to locally correct binary codes. Recall the definition of code concatenation.

Definition 1.7. Code Concetation For $q \geq 2, k \geq 1$ and $Q = q^k$, consider two codes which we call outer code and inner code:

$$\begin{aligned} C_{\text{out}} : [Q]^K &\rightarrow [Q]^N, \\ C_{\text{in}} : [q]^k &\rightarrow [q]^n. \end{aligned}$$

Note that the alphabet size of C_{out} exactly matches the number of messages for C_{in} , which means that we can have a bijection between $[Q]$ and $[q]^k$ (this means we can use C_{in} to encode any symbol in a codeword in C_{out}). Then given $m = (m_1, \dots, m_K) \in [Q]^K$, we have the code $C_{\text{out}} \circ C_{\text{in}} : [q]^{kK} \rightarrow [q]^{nN}$ defined as

$$C_{\text{out}} \circ C_{\text{in}}(m) = (C_{\text{in}}(C_{\text{out}}(m)_1), \dots, C_{\text{in}}(C_{\text{out}}(m)_N)),$$

where

$$C_{\text{out}}(m) = (C_{\text{out}}(m)_1, \dots, C_{\text{out}}(m)_N).$$

The main proposition of this section is:

Proposition 1.8. *Let $0 < \sigma < 1$, n and d be positive integers. Let $q = 2^b$ be a power of two such that $d \leq \sigma(q - 1) - 1$. Suppose that there exists a binary linear code C_{in} of distance μB encoding b -bit messages to B -bit codewords; then there exists a linear code C of dimension $k = \binom{n+d}{d} \cdot b$ in $\mathbb{F}_2^N$, $N = q^n \cdot B$, that for all positive $\delta < (1/2 - \sigma)\mu$ is $((q - 1)B, \delta, O_{\sigma, \mu, \delta}(1/q))$ -locally correctable.*

Here we take the code to be the code concetation of C_{in} and the code we discussed in the previous section, with C_{in} as the inner code. Then we want to find the inetrpolating polynomial h of degree atmost $2d$ by seeing agreement of h and C_{in} encoding of h by atmost $(1 - 2\sigma)(q - 1)\mu B/2$ queries.

Proof of failure probability. We want to decompose the same random variable X by weighted sum of the same indicator random variables as in 1.1. For $\mathbf{a} \in \mathbb{F}_q^n$, define $t_{\mathbf{a}}$ to be the number

of corruptions in the C_{in} encoding of $\text{Eval}_{\mathbf{a}}(f)$, so we have $\sum_{\mathbf{a} \in \mathbb{F}_q^n} t_{\mathbf{a}} \leq \delta B q^n$. Now for $\lambda \in \mathbb{F}_q^*$,

$$X^\lambda = \sum_{\mathbf{a} \in \mathbb{F}_q^n} t_{\mathbf{a}} \cdot \mathbb{1}_{\{\chi(\lambda) = \mathbf{a}\}}.$$

So total number of corrupted query points is $X = \sum_{\lambda \in \mathbb{F}_q^*} X^\lambda$. Now the failure probability that we want to calculate is

$$\Pr \left[X \geq \frac{1}{2}(1 - 2\sigma)(q - 1)\mu B \right].$$

Now similarly we get that $\mathbb{E}[X] \leq \delta(q - 1)B$ and $\mathbb{D}[X] \leq (\delta - \delta^2)(q - 1)B^2$, hence applying Chebyshev, we get out desired bound for the failure probability. $\square$

CHAPTER 2

MATCHING VECTOR CODES

2.1 Introduction

In the RM codes seen in Chapter 1, we could do local correction upto $\delta < 1/4$ fraction of error tolerance. Our main goal of this chapter would be to construct a code that can tolerate upto $\delta < 1/2$ fraction of errors, which would be optimal for unique local decoding.

We first define what *matching vectors* are, then we can specify the encoding of *matching vector code*.

Definition 2.1 (Matching Vectors). Let $S \subseteq \mathbb{Z}_m^n$, we say that the families $\mathcal{U} = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ and $\mathcal{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ of vectors in $\mathbb{Z}_m^n$ forms an *S-matching family* if

1. For all $i \in [k]$, $(\mathbf{u}_i, \mathbf{v}_i) = 0$,
2. For all $i \neq j$ in $[k]$, $(\mathbf{u}_j, \mathbf{u}_i) \in S$.

Observation 2.2. Let g be a generator of the unique subgroup C_m of $\mathbb{F}_q^*$ of size m . Then for any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{Z}_m^n$ and $\lambda \in \mathbb{Z}_m$,

$$\begin{aligned} \text{mon}_{\mathbf{u}}(g^{w+\lambda v}) &= \prod_{l=1}^n \left(g^{\mathbf{w}^{(l)} + \lambda \mathbf{v}^{(l)}} \right)^{\mathbf{u}^{(l)}} \\ &= \prod_{l=1}^n g^{\mathbf{w}^{(l)} \mathbf{u}^{(l)}} \cdot \prod_{l=1}^n (g^{\lambda})^{\mathbf{v}^{(l)} \mathbf{u}^{(l)}} \\ &= g^{(\mathbf{u}, \mathbf{w})} \cdot (g^{\lambda})^{(\mathbf{v}, \mathbf{u})}. \end{aligned}$$

So we can see that $M_{\mathbf{w}, \mathbf{v}}$ -evaluation of $\text{mon}_{\mathbf{u}}$ is a C_m -evaluation of a univariate polynomial

$$g^{(\mathbf{u}, \mathbf{w})} \cdot y^{(\mathbf{v}, \mathbf{u})} \in \mathbb{F}_q[y].$$

Now we define the encoding of matching vector codes, but we first need to assume that there exists an S -matching family $\mathcal{U}$ and $\mathcal{V}$ of size k . We shall deal with this existence later. There is also an abstract decoding method mentioned that will be the main motivation behind all the decoding procedure in the coming sections.

(ENCODING) We take a k -length message $\mathbf{x} = (x(1), \dots, x(k)) \in \mathbb{F}_q^k$ and encode it by C_m^n -evaluations of the polynomial

$$F_{\mathbf{x}}(z_1, \dots, z_n) = \sum_{j=1}^k x(j) \text{mon}_{u_j}(z_1, \dots, z_n). \quad (2.1)$$

(ABSTRACT DECODING) We want to retrieve $x(i)$ for a specified index $i \in [k]$. The decoding procedure is:

1. Pick $\mathbf{w} \sim \text{Unif}(\mathbb{Z}_m^n)$.
2. Compute the restriction of $F \equiv F_{\mathbf{x}}$ to $M_{\mathbf{w}, v_i}$ by some queries of the corrupted M_{w, v_i} -evaluation of F at less than or equal to m points.

One might ask now that what is the importance of this restriction and how it will help in retrieving $x(i)$. The following observation gives us the motivation of using matching family and this second step.

Observation 2.3. The M_{w, v_i} -evaluation of F is equivalent to the C_m^n -evaluation of the polynomial f defined as:

$$f(y) = \sum_{j=1}^k x(j) g^{(\mathbf{u}_j, \mathbf{w})} \cdot y^{(\mathbf{v}_i, \mathbf{u}_j)}$$

By matching family property, we can write the polynomial as:

$$f(y) = x(i) g^{(\mathbf{u}_i, \mathbf{v}_i)} + \sum_{s \in S} \left(\sum_{j: (\mathbf{u}_j, \mathbf{v}_i) = s} x(j) g^{(\mathbf{u}_j, \mathbf{v}_i)} \right) y^s.$$

So we can get $x(i)$ by just computing $f(0) / g^{(\mathbf{u}_i, \mathbf{w})}$. But of course we can not construct this

polynomial, so we try to find the restriction of F to $M_{\mathbf{w}, \mathbf{v}_i}$.

2.2 Some Decoding Procedures

We shall now see some explicit decoding procedures along with making the error tolerance better and better.

2.2.1 Basic Decoding on Lines

Proposition 2.4. *Let $\mathcal{U}, \mathcal{V}$ be a family of S -matching vectors in $\mathbb{Z}_m^n$, with $|\mathcal{U}| = |\mathcal{V}| = k$ and $|S| = s$. Suppose $m \mid (q - 1)$, where q is a prime power. Then there exists a q -ary linear code encoding k -long messages to m^n -long codewords that is $(s + 1, \delta, (s + 1)\delta)$ -locally decodable for all δ .*

Consider the code specified in Equation 2.1. Now do the following decoding procedure:

1. Pick $\mathbf{w} \sim \text{Unif}(\mathbb{Z}_m^n)$ and construct the multiplicative line $M_{\mathbf{w}, \mathbf{v}_i}$.
2. Query the M -evaluation of F at $s + 1$ many consecutive locations, say $\{g^{\mathbf{w} + \lambda \mathbf{v}_i} : \lambda \in \{0, \dots, s\}\}$ and obtain the points $c_0, \dots, c_s$.
3. Interpolate the unique sparse polynomial $h \in \mathbb{F}_q[y]$ with $\text{supp}(h) \subseteq S \cup \{0\}$ such that $h(g^\lambda) = c_\lambda$ for all λ .
4. Output $h(0)/g^{(\mathbf{w}, \mathbf{u}_i)}$.

Proof for failure probability. This is a fairly easy probability to calculate.

$$\begin{aligned}
 \Pr[\text{failure of decoder}] &= \Pr[F(g^{\mathbf{w} + \lambda \mathbf{v}_i}) \neq c_\lambda \text{ for some } \lambda] \\
 &\leq \sum_{\lambda} \Pr[F(g^{\mathbf{w} + \lambda \mathbf{v}_i}) \neq c_\lambda] \\
 &\leq \sum_{\lambda} \delta = (s + 1)\delta.
 \end{aligned}$$

□

2.2.2 Improvement on Line Decoding

In this section we shall see how we can make the line decoding better to get error tolerance of $\delta < 1/4$ by making the matching families *bounded*.

Definition 2.5 (*b*-bounded Matching Vectors). $\mathcal{U}, \mathcal{V}$ are said to be a *b*-bounded *S*-matching family over $\mathbb{Z}_m^n$ if *S* is *b*-bounded.

We now give the main proposition of this subsection analogous to Proposition 1.5.

Proposition 2.6. *Let $0 < \sigma < 1$. Let $\mathcal{U}, \mathcal{V}$ be a σm -bounded family of *S*-matching vectors in $\mathbb{Z}_m^n$, $|\mathcal{U}| = |\mathcal{V}| = k$. Suppose $m \mid (q - 1)$, where q is a prime power. Then there exists a q -ary linear code encoding k -long messages to m^n -long codewords that is $(m, \delta, 2\delta / (1 - \sigma))$ -locally decodable for all δ .*

Consider the same code as before and the following decoding procedure:

1. Pick $\mathbf{w} \sim \text{Unif}(\mathbb{Z}_m^n)$ and construct the multiplicative line $M_{\mathbf{w}, \mathbf{v}_i}$.
2. Query the *M*-evaluation of *F* at all of the m locations, and obtain the points $c_0, \dots, c_{m-1}$.
3. By Berlekamp-Welch algorithm, given in [GRS25], compute the unique univariate polynomial $h(y) \in \mathbb{F}_q[y]$ of degree less than σm and $\text{supp}(h) \subseteq S \cup \{0\}$ such that *h* agrees with the corrupted *M*-evaluation of *F* at all but at most $m(1 - \sigma)/2$ many points.
4. Return $h(0)/g^{(\mathbf{u}_i, \mathbf{w})}$ if such *h* exists, else return 0.

Proof for failure probability. Take the random variable *X* denoting the number of corrupted query points in the *M*-evaluation of *F*. Now failure occurs if $X \geq m(1 - \sigma)/2$, hence failure probability is less than or equal to:

$$\begin{aligned}
 \Pr \left[X \geq \frac{m(1 - \sigma)}{2} \right] &\leq \frac{1}{m(1 - \sigma)/2} \mathbb{E}[X] \quad [\text{Markov Inequality}] \\
 &= \frac{1}{m(1 - \sigma)/2} \sum_{\lambda \in \mathbb{Z}_m} \Pr \left[F(g^{\mathbf{w} + \lambda \mathbf{v}_i}) \neq c_\lambda \right] \\
 &\leq \frac{2m\delta}{m(1 - \sigma)} = \frac{2\delta}{1 - \sigma}.
 \end{aligned}$$

□

Observation 2.7 (Error Tolerance). As $\sigma \downarrow 0$, the limiting failure probability is 2δ , hence we can only tolerate up to $\delta < 1/4$ fraction of errors, because for error tolerance, the failure probability must be less than $1/2$. In the next subsection, we show a decoding paradigm that can tolerate up to $1/2$ fraction of errors.

2.2.3 Decoding on Multiple Lines

Rather than constructing one random multiplicative line, we shall construct more and using the queries at each M -evaluation of F , we shall interpolate the polynomial.

Proposition 2.8. *Let $0 < \sigma < 1$. Let $\mathcal{U}, \mathcal{V}$ be a σm -bounded family of S -matching vectors in $\mathbb{Z}_m^n$, $|\mathcal{U}| = |\mathcal{V}| = k$. Suppose $m \mid (q - 1)$, where q is a prime power. Then for every positive integer l there exists a q -ary linear code encoding k -long messages to m^n -long codewords that for all $\delta < (1 - \sigma)/2$ is $(lm, \delta, \exp_{\delta, \sigma}(-l))$ -locally decodable.*

Consider the same code as before and fix an $l \in \mathbb{N}$. Now here we shall not construct the polynomial like we have done before, we shall consider l many M -evaluations of F and instead of interpolating the polynomial directly, we shall assign weight to each field element e with respect to the likelihood of $x(i) = e$. And from that we shall return the unique highest weight field element, if exists. The decoding procedure is the following:

1. Take $\mathbf{w}_1, \dots, \mathbf{w}_l$ uniformly at random from $\mathbb{Z}_m^n$ and consider the l many multiplicative lines $\{M_{\mathbf{w}_j, \mathbf{v}_i} : j \in [l]\}$.
2. Query at the lm many coordinates of all the M -evaluations of F and obtain l many vectors in $\mathbb{F}_q^m$, say $\mathbf{c}^1, \dots, \mathbf{c}^m$. For each $e \in \mathbb{F}_q$, compute:

$$\text{weight}(e) = \sum_{j=1}^l \text{weight}(e, M_{\mathbf{w}_j, \mathbf{v}_i}).$$

3. Return e with the highest weight. If it is not unique, then return 0.

Observation 2.9. Suppose there are two distinct $e_1, e_2 \in \mathbb{F}_q$ such that $\text{weight}(e_1), \text{weight}(e_2) >$

$lm(1 + \sigma)/2$. Then by PHP, there would be two distinct polynomials $h_1, h_2 \in \mathbb{F}_q[y]$, both of degree less than σm and two multiplicative lines M_1, M_2 such that $\text{weight}(e_1, M_1) > m(1 + \sigma)/2$ and $\text{weight}(e_2, M_2) > m(1 + \sigma)/2$. Now by inclusion-exclusion, the number of agreement points of h_1 and h_2 is at least $2m(1 + \sigma)/2 - m = \sigma m$. Thus we can not have such a pair.

Proof for failure probability. Consider the random variable X denoting the number of corrupted queried points and failure probability is less than or equal to $\Pr[X \geq lm(1 + \sigma)/2]$. Note that,

$$X = \sum_{j=1}^l \left(\sum_{\lambda \in \mathbb{Z}_m} \mathbb{1}_{\{F(g_\lambda^{(\mathbf{w}_j, \mathbf{v}_i)}) \neq \mathbf{c}_\lambda^j\}} \right)$$

Now the expected value of X is:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{j=1}^l \left(\sum_{\lambda \in \mathbb{Z}_m} \Pr[F(g_\lambda^{(\mathbf{w}_j, \mathbf{v}_i)}) \neq \mathbf{c}_\lambda^j] \right) \\ &\leq \sum_{j=1}^l \left(\sum_{\lambda \in \mathbb{Z}_m} \delta \right) = ml\delta. \end{aligned}$$

Now X is the sum of lm many independent and identically distributed Bernoulli random variable of success probability δ . Now we can apply *Chernoff bound*, given in [AS00], in the following way to find an upper bound for failure probability. Using $\delta < (1 - \sigma)/2$,

$$\begin{aligned} \Pr[X \geq lm(1 + \sigma)/2] &\leq \Pr[X - \mathbb{E}[X] \geq lm(1 + \sigma)/2 - lm(1 - \sigma)/2] \\ &= \Pr[X - \mathbb{E}[X] \geq ml\sigma] \\ &\leq \exp_{\sigma, \delta}(-l) \quad \text{Chernoff Bound.} \end{aligned}$$

□

Now we can show how we increase the error tolerance up to $\delta < 1/2$. Since the proposition holds for every $\delta < (1 - \sigma)/2$, letting $\sigma \downarrow 0$ yields $\delta \uparrow 1/2$. Hence the decoder tolerates any error fraction $\delta < 1/2$.

CHAPTER 3

MATCHING VECTORS

Recall the definition of *matching vectors*, given in Definition 2.1. We shall give some combinatorial and algorithmic viewpoint for constructing such matching vector families in this chapter. The main references for this are [MBBR92] and [Gro00].

This chapter shall mainly consist of the construction of an MVF. The main result of [Gro00] is the following:

Theorem 3.1 (Grolmusz). *Let $m = \prod_{i=1}^t p_i$ be a product of distinct primes. Then there exists an explicit matching vector family in $\mathbb{Z}_m^l$ of size*

$$k = \exp \left(\Omega \left(\frac{(\log l)^t}{(\log \log l)^{t-1}} \right) \right).$$

3.1 Degree Analysis of OR_n Function

Before going to the actual proof of Theorem 3.1, we first state some interesting ideas. One of the most important combinatorial question that can arise in one's mind is how large the size of the family can be. The following lemma gives an upper bound on the size of an MVF.

Lemma 3.2. *If p is a prime, then any MVF in $\mathbb{Z}_p^l$ has size at most $1 + l^{p-1}$.*

Proof. Let $\mathcal{U}, \mathcal{V}$ be a MVF in $\mathbb{Z}_p^l$ of size k . Consider the matrix $A \in \mathbb{Z}_p^{k \times k}$ with $A_{i,j} = \langle \mathbf{u}_i, \mathbf{v}_j \rangle$. By Fermat's Little Theorem, Lemma A.3 in Appendix A, we observe that $A_{i,j}^{p-1} = 1$ for all i, j . Observe, $A^{p-1} = J_k - I_k$, hence $\text{rank}(A^{p-1}) \leq k - 1$ and $\text{rank}(A^{p-1}) \leq \text{rank}(A)^{p-1} \leq l^{p-1}$, by Lemma A.2. □

3.1.1 Polynomial Representation of OR-function in mod 6

Definition 3.3 (OR Function). n -variate OR function, denoted as OR_n , is a boolean function defined as:

$$\text{OR}_n(x_1, \dots, x_n) = 0 \text{ if and only if } x_i = 0 \text{ for all } i \in [n].$$

We shall now study some polynomial “representative”(s) of this OR function.

Definition 3.4. A polynomial $g \in \mathbb{Z}_m[x_1, \dots, x_n]$ represents a boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ mod m if for all $\mathbf{x} \in \{0, 1\}^n$,

$$g(\mathbf{x}) \begin{cases} = 0 \pmod{m} & \text{if } f(\mathbf{x}) = 0 \\ \neq 0 \pmod{m} & \text{if } f(\mathbf{x}) = 1 \end{cases}$$

And the *degree* of the boolean function f modulo p , denoted as $\deg_p(f)$, is defined as $\min\{\deg(g) : g \text{ represents } f \text{ modulo } p\}$.

The following lemma shall give the main motivation behind the study of representative polynomials modulo a composite number with at least two distinct prime factors, instead of a prime power, or equivalently, shifting our focus from fields to rings!

Lemma 3.5. *If p is a prime, then $\deg_p(\text{OR}_n) \geq n/(p-1)$.*

Proof. Let g be a polynomial representative of OR_n in $\mathbb{F}_p$, i.e., $\text{OR}_n(\mathbf{a}) = 0$ if and only if $g(\mathbf{a}) = 0 \pmod{p}$. Notice that as we are working $\mathbb{F}_p$, we can transform g to make it behave like a boolean function in $\mathbb{F}_p$. Consider the polynomial $h(\mathbf{x}) = 1 - g^{p-1}(\mathbf{x})$. Now notice that as we are in $\{0, 1\}^n$, we can make this h multilinear, say $\tilde{h}$. Observe, by Lemma A.3, $\tilde{h}(\mathbf{a}) = 1$ at $\mathbf{0}$ and 0 everywhere else. Also by Lemma A.6, we get that:

$$2^n - 1 = |\{\mathbf{a} \in \{0, 1\}^n : \tilde{h}(\mathbf{a}) = 0\}| \leq 2^n - 2^{n-\deg \tilde{h}}.$$

Hence, $\deg \tilde{h} \geq n$ and with that we get our final bound:

$$n \leq \deg \tilde{h} \leq \deg h \leq (p-1) \deg g.$$

□

Also note we can extend Lemma 3.5 to prime powers, the only difference that would rise it taken care by Lemma A.3. Hence, we see that taking prime power ring, or the field, we get the minimum degree to be $\Omega(n)$ but [MBBR92] showed that for *product of distinct primes* size ring, we can get much smaller degree representative polynomials. We shall only see the proof of BBR theorem for modulo 6 but this can be extended to any m that is a product of distinct primes with the same principles used in the following proof.

3.1.2 Barrington-Beigel-Rudich Theorem

In this subsection, we shall show that using $\mathbb{Z}_6$ as the base ring for the polynomials, we can reduce the degree of OR_n function to $O(\sqrt{n})$.

Theorem 3.6 (Barrington-Beigel-Rudich). *There exists an n -variate polynomial that represents OR_n modulo 6 with degree at most $O(\sqrt{n})$.*

Before going to this involved proof, let me give an intuitive proof outline of this theorem. Basically, we want to construct a polynomial $f \in \mathbb{Z}_6[x_1, \dots, x_n]$ of degree at most $O(\sqrt{n})$ such that $\text{OR}_n(\mathbf{a}) = 0$ if and only if $f(\mathbf{a}) = 0 \pmod{6}$. Suppose we can construct two polynomials $f_2 \in \mathbb{Z}_2[x_1, \dots, x_n]$ and $f_3 \in \mathbb{Z}_3[x_1, \dots, x_n]$ such that for some $2^d \approx \sqrt{n}$ and $3^e \approx \sqrt{n}$, we have:

$$f_2(a) = 0 \iff \|a\| \equiv 0 \pmod{2^d} \quad \text{and} \quad f_3(a) = 0 \iff \|a\| \equiv 0 \pmod{3^e},$$

making sure that $\deg f_i$ is nearly $\sqrt{n}$. Intuitively this is possible due to the fact that the number of $\mathbf{a}$ that satisfies the above condition is at most $\sum_{i=1}^{\sqrt{n}} \binom{n}{i\sqrt{n}}$, which is very smaller than $\sqrt{n}2^{n-1}$, which is the *Schwarz-Zippel* bound, see Lemma A.5. Having this done, let us consider the isomorphism given in Theorem A.8. There exists a unique polynomial $f \in \mathbb{Z}_6[x_1, \dots, x_n]$ such that $f_2 = f \pmod{2}$ and $f_3 = f \pmod{3}$. If we make sure that $2^d 3^e > n$,

then we can show that f represents OR_n by the following arguments and Theorem A.7.

$$f(\mathbf{a}) = 0 \iff f_2(\mathbf{a}) = 0 \text{ and } f_3(\mathbf{a}) = 0 \iff \|\mathbf{a}\| = 0 \pmod{2^d 3^e} \iff \mathbf{a} = \mathbf{0},$$

because $\|\mathbf{a}\| \leq n$. And to see the degree of f , note that $\deg f \leq \max\{\deg f_2, \deg f_3\} = O(\sqrt{n})$.

Lemma 3.7. *If p is fixed prime, then for all $d \in \mathbb{N}$, there exists a polynomial $f_p^d \in \mathbb{Z}_p[x_1, \dots, x_n]$ such that $\deg(f_p^d) \leq p^d - 1$ and $f_p^d(\mathbf{a}) = 0$ if and only if $\|\mathbf{a}\| = 0 \pmod{p^d}$.*

Proof. We shall use Theorem A.8 to show this. Let $w = \|\mathbf{a}\|$ with $(w_0, w_1, \dots)$ as the p -ary expansion of w . Consider the function $f(\mathbf{a}) = \prod_i (1 - w_i^{p-1}) - 1 \pmod{p}$, so that $f(\mathbf{a}) = 0$ if and only if $\mathbf{a} = \mathbf{0}$, this is because $f(\mathbf{a}) = 0$ if and only if $w_i = 0$ for all i . Now the next question is if we can write this $w_j(\mathbf{a})$ as a polynomial in $\mathbf{a}_1, \dots, \mathbf{a}_n$. Note that by Theorem A.4 for all j ,

$$\begin{aligned} \binom{w}{p^j} &\equiv \binom{w_j}{1} \prod_{r \neq j} \binom{w_r}{0} \pmod{p} \\ &= w_j. \end{aligned}$$

So we can re-write the function as $f(\mathbf{a}) = \prod_i \left(1 - \binom{w}{p^i}^{p-1}\right) - 1 \pmod{p}$. And one possible candidate for this $\binom{w}{p^i}$ is the multilinear polynomial $\text{Sym}_{p^i}(\mathbf{x})$ defined for any $\mathbf{x} \in \mathbb{Z}_p$ as:

$$\text{Sym}_{p^i}(\mathbf{x}) = \sum_{\substack{I \subseteq [n] \\ |I|=p^i}} \prod_{j \in I} x_j.$$

Now observe that $\text{Sym}(\mathbf{a}) = \binom{w}{p^i}$, thus one possible candidate polynomial in $\mathbb{Z}_p[x_1, \dots, x_n]$ is:

$$f_p^d(\mathbf{x}) = \prod_{i=0}^{d-1} \left(1 - \text{Sym}_{p^i}(\mathbf{x})^{p-1}\right) - 1.$$

Degree of this polynomial is at most $\sum_{j=0}^{d-1} p^j(p-1) = p^d - 1$ and it represents OR_n . $\square$

Now we can prove Theorem 3.6 formally.

Proof of Theorem 3.6. Take d and e such that $\sqrt{n} < 2^d \leq 2\sqrt{n}$ and $\sqrt{n} < 3^e \leq 3\sqrt{n}$. By Lemma 3.7, there are two polynomials $f_2 \in \mathbb{Z}_2[x_1, \dots, x_n]$ and $f_3 \in \mathbb{Z}_3[x_1, \dots, x_n]$ such that $\deg(f_2) \leq 2^d - 1$, $\deg(f_3) \leq 3^e - 1$, and

$$f_2(\mathbf{a}) = 0 \iff \|\mathbf{a}\| \equiv 0 \pmod{2^d}, \quad f_3(\mathbf{a}) = 0 \iff \|\mathbf{a}\| \equiv 0 \pmod{3^e}.$$

Now by Theorem A.8, $f = (f_2, f_3) \in \mathbb{Z}_6[x_1, \dots, x_n]$ has degree at most $\max(\deg(f_2), \deg(f_3))$, which is at most $3\sqrt{n}$, and $f(\mathbf{a}) = 0$ if and only if $\|\mathbf{a}\| \equiv 0 \pmod{2^d 3^e}$. Since $2^d 3^e > n$, we have $\|\mathbf{a}\| \equiv 0 \pmod{2^d 3^e}$ if and only if $\|\mathbf{a}\| = 0$ if and only if for all $i \in [n]$, $\mathbf{a}_i = 0$. So $f(\mathbf{a}) = 0$ if and only if for all $i \in [n]$, $\mathbf{a}_i = 0$. $\square$

3.2 Construction of MVF

Now we have built pretty much all the machinery needed to prove Theorem 3.6. The version of this theorem for modulo 6 is the following:

Theorem 3.8. *For any $l \in \mathbb{N}$, there exists a matching vector family in $\mathbb{Z}_6^l$ of size $l^{C \log l / (\log \log l)^2}$, where $C = 1/81$.*

Proof. Start with an arbitrary $h \in \mathbb{N}$ that will be decided later precisely. Now take two boolean strings $\mathbf{a}$ and $\mathbf{b}$ of size h . By Theorem 3.8, there exists a polynomial in $\mathbb{Z}_6[x_1, \dots, x_h]$ that represents OR_h modulo 6 of degree $d = O(\sqrt{n})$. For a fixed $\mathbf{b}$, define $g_{\mathbf{b}} \in \mathbb{Z}_6[x_1, \dots, x_h]$ such that $g_{\mathbf{b}}(\mathbf{x}) = f((\mathbf{b} - \mathbf{x}) \odot (\mathbf{b} - \mathbf{x}))$. Observe that,

$$g_{\mathbf{b}}(\mathbf{a}) \begin{cases} = 0 \pmod{6} & \text{if } \mathbf{a} = \mathbf{b} \\ \neq 0 \pmod{6} & \text{if } \mathbf{a} \neq \mathbf{b}. \end{cases}$$

We shall now try to construct a matching vector with these 2^h many kind of matching vector family of $\mathcal{G} = \{g_{\mathbf{b}}\}_{\mathbf{b} \in \{0,1\}^h}$ and $\mathcal{A} = \{\mathbf{a}\}_{\mathbf{a} \in \{0,1\}^h}$. The following lemma helps us con-

structing an MVF in $\mathbb{Z}_6^l$ of size $\binom{h+d}{d}$.

Lemma 3.9. *$\mathcal{G}$ and $\mathcal{A}$ yields $\mathbb{Z}_6$ -matching in $\mathbb{Z}_6^{\binom{h+d}{d}}$ of size 2^h .*

Proof. We define the vector $\mathbf{u}_b$ to be the $\binom{h+d}{d}$ -dimensional vector of coefficients of the polynomial g_b . Similarly, for $\mathbf{a} \in \{0, 1\}^h$, we define the vector $\mathbf{v}_a$ to be the vector of evaluations of monomials at the point $\mathbf{a}$. It is easy to check that for all $\mathbf{a}, \mathbf{b} \in \{0, 1\}^h$,

$$\langle \mathbf{u}_b, \mathbf{v}_a \rangle = g_b(\mathbf{a}),$$

and hence the sets U, V indeed form an $\mathbb{Z}_6$ -matching family. $\square$

Set h such that $l = \binom{h+d}{d} \leq (2h)^d \leq (2h)^{3\sqrt{h}}$. Now it is enough to show $h \geq C \log^2 l / \log^2 \log l$.

This follows directly from the following calculations:

$$\begin{aligned} \frac{C \log^2 l}{\log^2 \log l} &\leq \frac{\left(3\sqrt{h} \log(2h)\right)^2}{81 \log^2 \left(3\sqrt{h} \log(2h)\right)} \\ &\leq \frac{9h \log^2(2h)}{81 \log^2(\sqrt{h})} \\ &= \frac{h \left(2 \log(\sqrt{h}) + 1\right)^2}{9 \log^2(\sqrt{h})} \\ &\leq \frac{9h \log^2(\sqrt{h})}{9 \log^2(\sqrt{h})} \\ &= h \end{aligned}$$

$\square$

This proof can be generalized to modulo m for any $m (= \prod_i^t p_i)$ which is a product of distinct primes by setting the prime powers $p_i^{e_i}$ used in the construction to be greater than $n^{1/t}$ and at most $p_i n^{1/t}$ and use the generalized Chinese Remainder Theorem.

CHAPTER 4

PRIVATE INFORMATION RETRIEVAL

Private Information Retrieval schemes were first introduced in the paper [CKGS98]. This chapter will consist of the PIR scheme described in [GKS25].

4.1 Introduction

In a k -server Private Information Retrieval (PIR) scheme, a user wishes to retrieve the i -th bit a_i of an n -bit database $\mathbf{a} = (a_1, \dots, a_n) \in \{0, 1\}^n$, which is replicated among k non-communicating servers, without revealing any information about the index i to any individual server. The central goal in the study of PIR schemes is to minimize the *communication cost*, defined as the worst-case total number of bits exchanged between the user and the servers over the course of the protocol. The trivial solution, which works even with a single server, is to have the server transmit the entire database $\mathbf{a}$ to the user; this achieves correctness and perfect privacy, but at the high cost of n bits of communication. A fundamental question is therefore: *can we do better?*

Before moving forward let me define formally what a PIR scheme is:

Definition 4.1 (PIR Scheme). A one-round t -server PIR protocol involves t servers $S_1, \dots, S_t$, each holding the same n -bit database $\mathbf{a} = (a_1, \dots, a_n) \in \{0, 1\}^n$, such that these servers do not communicate with each other, and there is a user $\mathcal{U}$ who knows n and wishes to retrieve some bit a_i for $i \in [n]$, without revealing the index i . The protocol consists of a randomized algorithm for the user and t deterministic algorithms for the servers, satisfying the following:

1. On input $i \in [n]$, the user $\mathcal{U}$ samples a random string rand , produces t queries $q_1, \dots, q_t$,

and sends query q_j to server S_j .

2. Each server S_j computes a response $r_j = S_j(\mathbf{a}, q_j)$ and sends it back to the user.
3. The user, given i , rand , and responses $r_1, \dots, r_t$, reconstructs a_i .

The protocol must satisfy the following two conditions:

- **Correctness:** For any database $\mathbf{a} \in \{0,1\}^n$ and any index $i \in [n]$, the user outputs a_i with probability 1, where the probability is taken over the random string rand .
- **Privacy:** Each server individually learns nothing about the queried index i . Formally, for any server S_j and any two indices $i_1, i_2 \in [n]$, the distributions of $q_j(i_1, \text{rand})$ and $q_j(i_2, \text{rand})$ over the randomness rand are identical.

For 1-server PIR scheme, we can not do better than n communication cost as if we have less than n bits of communication, then there would be two indices for which we may have the same message from the server. In [CKGS98], we have a 2-server PIR scheme with $\mathcal{O}(n^{1/3})$ and 3-server PIR scheme with $\mathcal{O}(n^{1/2})$ communication cost. Then in [Efr09], we have a communication $2^{\mathcal{O}(\sqrt{\log n})}$ for 3-server PIR scheme.

4.2 A Basic PIR Scheme

The following PIR Scheme will give us a $\mathcal{O}(n^{1/2})$ communication with 3 servers. If $\mathbf{a} \in \mathbb{F}^n \equiv \mathbb{F}^{\binom{k}{2}}$ is our database and user wants $\mathbf{a}_\tau$, then it equivalent to construct a polynomial $F \in \mathbb{F}[x_1, \dots, x_k]$ defined as:

$$F(x_1, \dots, x_k) = \sum_{i,j} a_{ij} X_i X_j.$$

For some re-design of the indexing set and have the user find a suitable a_{ij} such that it equals a_τ , by giving the evaluation of F at $(0, \dots, 1, \dots, 1, \dots, 0)$, with 1 only appearing in i and j -th coordinates.

We generalize this problem in the given PIR scheme and solve it via 3-servers and $\mathcal{O}(n^{1/2})$ communication.

PIR Scheme

Suppose for an $\mathbf{x} \in \mathbb{F}^k$, the user wants $F(\mathbf{x})$ and the servers has the polynomial F .

1. Pick $\mathbf{y} \in \mathbb{F}^k$ uniformly at random.
2. Consider the curve $C(Z) = \mathbf{x} + Z\mathbf{y}$.
3. Consider the restriction $A(Z) = F(C(Z)) = F(\mathbf{x} + Z\mathbf{y})$ of F on this curve, notice that $\deg A = 2$.
4. For $i \in [3]$, ask server i to return $A(i)$ to the user.
5. Interpolate A using $A(1), A(2)$ and $A(3)$.
6. Return $A(0)$.

The correctness of the PIR scheme is easy to observe. And we send k -bit information to each server and receive 1-bit information from each server. Thus in total the communication cost is $\mathcal{O}(k) = \mathcal{O}(n^{1/2})$ as $n = \binom{k}{2}$.

4.3 PIR Scheme of [GKS25]

The PIR scheme described in Section 4.2 is the main idea behind all the PIR schemes that are to come. We pick a random curve in a space and restrict the polynomial(not exactly the same as in Section 4.2) to that curve and try to interpolate the polynomial, if the entire polynomial can not be interpolated(which will be the case), then we shall try to interpolate the value of the polynomial at 0 only.

PIR schemes using matching vectors were introduced in [Yek08] and [Efr09]. A very detailed and well-written flow of improvement of the PIR schemes throughout the years can be found in [GKS25]. But we shall only see and explore the improvement done by the very paper.

But before going to the actual explicit PIR scheme, we need to be familiar with some ideas required for the scheme to make sense and implement.

Suppose the database here is $\mathbf{a} \in \mathbb{F}^n$ and the user wants a_τ . Also $(\mathbf{u}_1, \mathbf{v}_1), \dots, (\mathbf{u}_n, \mathbf{v}_n)$ is an S -matching vector family in $\mathbb{Z}_m^k$ for some positive integer m . Now the servers make the polynomial $F(\mathbf{X}) \in \mathbb{F}[X_1, \dots, X_k]$ as follows:

$$F(\mathbf{X}) = \sum_{i=1}^n a_i \mathbf{X}^{\mathbf{u}_i} = \sum_{i=1}^n a_i X_1^{u_{i1}} \cdots X_k^{u_{ik}}.$$

We shall deal with the restriction of this polynomial throughout the chapter. Also we choose the characteristic p field $\mathbb{F}$ such that it contains all the m -th root of unity.

We first construct the random curve $C(Z)$ by first choosing β uniformly at random and considering $C(Z) = \beta \odot Z^{\mathbf{v}_\tau}$. Then consider the univariate polynomial $A(Z) = F(C(Z))$, but $\deg(A) = \mathcal{O}(k \cdot m^2)$. So to reduce it, we do $A_1(Z) = A(Z) \pmod{Z^m - 1}$. Observe that A_1 is of the form:

$$A_1(Z) = a_\tau \beta^{\mathbf{u}_\tau} + \sum_{\substack{s \in S \\ s \neq 0}} w_s Z^s.$$

So to get a_τ , it is enough to know $A_1(0)$. We shall try to interpolate A_1 at 0 in the following sections.

4.3.1 Preliminaries

We shall be doing the interpolation in the regime using *S-matching vector families* and *S-decoding polynomials*, or the SMVF+SSD framework. We have already seen SMVF before in Chapter 2, let us define what SSD is.

Definition 4.2 (*S-decoding Polynomials*). Let $S \subseteq \mathbb{Z}_m$ be a set containing 0. Let $\mathbb{F}$ be a field containing an element γ_m which is a primitive m 'th root of 1. A polynomial $P(Z) \in \mathbb{F}[Z]$ is called an *S-decoding polynomial* if the following conditions hold:

- $\forall s \in S \setminus \{0\} : P(\gamma_m^s) = 0$
- $P(\gamma_m^0) = P(1) = 1$

Now we can state the main theorem that is the result of [GKS25], which improves the communication and server trade-off.

Definition 4.3 (Canonical Set). Let $m \in \mathbb{N}$, we define *canonical set* for m to be:

$$S = \{x \in \mathbb{Z}_m : x^2 \equiv x \pmod{m}\}.$$

The following is the main result that we shall prove in the coming sections.

Theorem 4.4 ([GKS25]). Let m be a positive integer with r distinct prime factors, and let p be a prime not dividing m . Suppose $\mathbb{F}$ is a field of characteristic p such that, for $S^* \subseteq \mathbb{Z}_m$ being the canonical set for m , there is an S^* -decoding polynomial over $\mathbb{F}$ with sparsity $\leq t$. Then there is a t -server PIR scheme with communication $\exp\left(\tilde{O}\left((\log n)^{\frac{1}{r+1}}\right)\right)$.

This is a direct improvement of the result of [Efr09], which is:

Theorem 4.5 ([Efr09]). Let m be a positive integer with r distinct prime factors, and let p be a prime not dividing m . Suppose $\mathbb{F}$ is a field of characteristic p such that, for $S^* \subseteq \mathbb{Z}_m$ being the canonical set for m , there is an S^* -decoding polynomial over $\mathbb{F}$ with sparsity $\leq t$. Then there is a t -server PIR scheme with communication $\exp\left(\tilde{O}\left((\log n)^{\frac{1}{r}}\right)\right)$.

Following is the table of improvement over [Efr09] and [DG16] over communication of the form $\exp\left(\tilde{O}((\log n)^{1/c})\right)$ for positive integers c .

Total communication	# servers for [GKS25]	# servers for [DG16]	# servers for [Efr09]
$\exp(\tilde{O}((\log n)^{1/2}))$	2	2	3
$\exp(\tilde{O}((\log n)^{1/3}))$	3	4	8
$\exp(\tilde{O}((\log n)^{1/4}))$	8	8	9
$\exp(\tilde{O}((\log n)^{1/5}))$	9	16	24
$\exp(\tilde{O}((\log n)^{1/6}))$	24	32	27

Hasse Derivatives

Definition 4.6 (Hasse Derivatives). For a multivariate polynomial $F(\mathbf{X}) \in \mathbb{F}[\mathbf{X}]$ and a non-negative integer vector $\mathbf{i} = (i(1), \dots, i(k))$, we denote the $\mathbf{i}$ -th Hasse derivative of F by $F^{(\mathbf{i})}(\mathbf{X})$, and it is the coefficient of $\mathbf{Y}^{\mathbf{i}}$ in the polynomial $F(\mathbf{X} + \mathbf{Y}) \in \mathbb{F}[\mathbf{X}, \mathbf{Y}]$. Therefore, we have,

$$F(\mathbf{X} + \mathbf{Y}) = \sum_{\mathbf{i}} F^{(\mathbf{i})}(\mathbf{X}) \mathbf{Y}^{\mathbf{i}}.$$

We use the notation $F^{(<m)}(\mathbf{a})$ to be the vector of all the evaluations of the $\mathbf{i}$ -th Hasse derivative of F at point $\mathbf{a}$ for all vectors $\mathbf{i}$ such that $\text{wt}(\mathbf{i}) < m$. We note that Hasse derivatives are just a simple scaling of standard iterated derivatives whenever the order of derivative is smaller than the characteristic of the field. In particular, for the univariate polynomial $A(Z) = Z^s$, we have a simple formula for the Hasse derivative:

$$A^{(j)}(Z) = \binom{s}{j} Z^{s-j}.$$

Now the following facts about hasse derivatives are essential for interpolating $A_1()$ from its derivatives.

- Let $A(Z) \in \mathbb{F}[Z]$ be a univariate polynomial, and let $b \in \mathbb{F}$. Then knowing $A(Z) \bmod (Z - b)^m$ is exactly equivalent to knowing $A^{(<m)}(b)$. Concretely, given $A^{(<m)}(b)$ we can construct the polynomial

$$A^*(Z) = \sum_{i=0}^{m-1} A^{(i)}(b)(Z - b)^i,$$

which is easily seen to be the remainder of $A(Z) \bmod (Z - b)^m$. This connection goes in the other direction too.

- **Chain rule:** Suppose $F(X_1, \dots, X_k) \in \mathbb{F}[X_1, \dots, X_k]$ is a multivariate polynomial and $C = (P_1(Z), \dots, P_k(Z))$ is a tuple of univariate polynomials in the single variable Z . Denote their composition by the polynomial $A(Z) = F(C(Z))$. Then, the i -th Hasse derivative of A , $A^{(i)}$, can be expressed as a polynomial in $F^{(\mathbf{j})}(C(Z))$ for $\text{wt}(\mathbf{j}) \leq i$ and $P_t^{(j)}(Z)$ for $1 \leq t \leq k, j \leq i$.

4.3.2 0-interpolation Property

4.3.3 The PIR Scheme

We need the following things for the PIR scheme to be useful:

- $m \in \mathbb{N}$ that is a product of r distinct primes and a prime p not dividing m .

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APPENDIX A

APPENDIX

Lemma A.1. *If X, Y are two matrices over $\mathbb{F}_q$ of proper dimension, then $\text{rank}(X \otimes Y) = \text{rank}(X) \cdot \text{rank}(Y)$.*

Lemma A.2. *Suppose $k, t \in \mathbb{N}$ and $A, B \in \mathbb{F}_q^{k \times k}$ such that $B_{i,j} = A_{i,j}^t$ for all $i, j \in [k]$. Then $\text{rank}(B) \leq \text{rank}(A)^t$.*

Proof. Take the matrix $T = \otimes^t A$. Now notice that B is a sub-matrix of T . By Lemma A.1, $\text{rank}(B) \leq \text{rank}(T) = \text{rank}(A)^t$. $\square$

Lemma A.3 (Fermat's Little Theorem [Bur11]). *If $a \in \mathbb{Z}$ and $p \in \mathbb{P}$ such that $\gcd(a, p) = 1$, then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Theorem A.4 (Lucas' Theorem [Bur11]). *Let p be a prime and let m, n be non-negative integers with base- p representations $m = \sum_i m_i p^i$ and $n = \sum_i n_i p^i$. Then*

$$\binom{m}{n} \equiv \prod_i \binom{m_i}{n_i} \pmod{p}$$

Lemma A.5 (Schwarz–Zippel). *Let $f \in \mathbb{F}[x_1, \dots, x_n]$ be a non-zero polynomial of total degree d over a field $\mathbb{F}$, and let $S \subseteq \mathbb{F}$ be a finite set. If $r_1, \dots, r_n$ are chosen independently and uniformly at random from S , then:*

$$\Pr[f(r_1, \dots, r_n) = 0] \leq \frac{d}{|S|}.$$

Lemma A.6. *Suppose $f \in \mathbb{F}_p[x_1, \dots, x_n]$, where p is prime, is a multilinear non-zero polynomial of*

degree at most d , where $d > 0$. Then:

$$|\{a \in \{0,1\}^n \mid f(a) = 0\}| \leq 2^n - 2^{n-d},$$

where equality is achieved for $f = \prod_{i=1}^d x_i$.

Proof. We can prove the lemma by induction on n . In the base case, $n = 1$, we have $d = 1$, so f is an expression linear in x_1 , and thus can have at most one zero, so the inequality holds. Suppose we have shown the inequality holds for polynomials of $n - 1$ variables. We show that it also holds for any polynomial f of n variables. We can express f in the following way:

$$f(x_1, \dots, x_n) = x_1 g(x_2, \dots, x_n) + h(x_2, \dots, x_n).$$

Notice that the polynomial g is of degree at most $d - 1$. Then, we have:

$$\begin{aligned} |\{a \in \{0,1\}^n \mid f(a) = 0\}| &\leq 2|\{a \in \{0,1\}^{n-1} \mid g(a) = 0\}| + |\{a \in \{0,1\}^{n-1} \mid g(a) \neq 0\}| \\ &= |\{a \in \{0,1\}^{n-1} \mid g(a) = 0\}| + 2^{n-1} \\ &\leq 2^{n-1} - 2^{n-1-(d-1)} + 2^{n-1} \\ &= 2^n - 2^{n-d}, \end{aligned}$$

where the first inequality holds because if $g(x_2, \dots, x_n) \neq 0$, there is exactly one value of x_1 for which $f(x) = 0$, but if $g(x_2, \dots, x_n) = 0$, there might be up to two values of x_1 such that $f(x) = 0$. The second inequality holds by the inductive step. $\square$

Theorem A.7 (Chinese Remainder Theorem for Integer Ring). *Let p and q be prime numbers with $m = pq$. Then:*

$$\mathbb{Z}_m \cong \mathbb{Z}_p \times \mathbb{Z}_q,$$

where the isomorphism $\varphi : \mathbb{Z}_m \rightarrow \mathbb{Z}_p \times \mathbb{Z}_q$ is given by

$$\varphi(a) = (a \bmod p, a \bmod q).$$

Theorem A.8 (Chinese Remainder Theorem for Polynomial Rings). *Let p and q be prime numbers with $m = pq$. Then:*

$$\mathbb{Z}_m[x_1, \dots, x_n] \cong \mathbb{Z}_p[x_1, \dots, x_n] \times \mathbb{Z}_q[x_1, \dots, x_n],$$

where the isomorphism $\varphi : \mathbb{Z}_m[x_1, \dots, x_n] \rightarrow \mathbb{Z}_p[x_1, \dots, x_n] \times \mathbb{Z}_q[x_1, \dots, x_n]$ is given by

$$\varphi(f(x)) = (f(x) \bmod p, f(x) \bmod q).$$

This works because a polynomial in n variables modulo some number t is an expression built from sums and products, so φ can be applied coefficient-wise.